Evaluating, Transferring, and Ensembling

Modeling the distribution of biodiversity under global climate change

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Model Transferability & environmental novelty

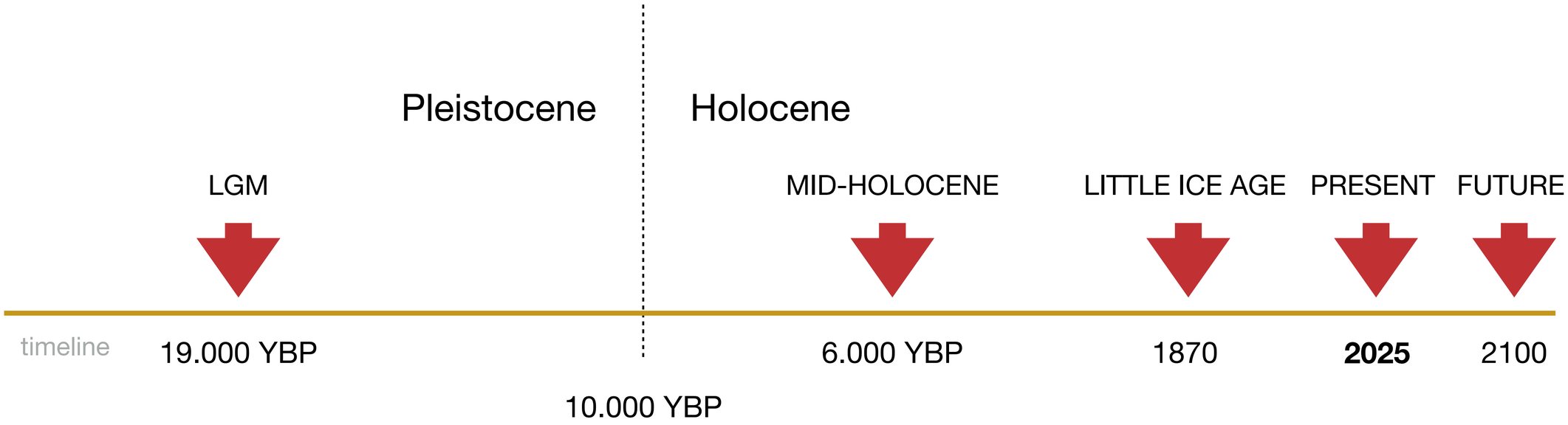
Model transferability

Ability of a model to predict beyond the data it was originally trained on.

Particularly **relevant when forecasting species distributions** under **climate change**.

Mainly dependent on:

1. Adequacy of environmental and species occurrence data to train the models;
2. Reduced complexity of the models (e.g., number of variables);
2. **Proper choice / tuning of model parameterization.**



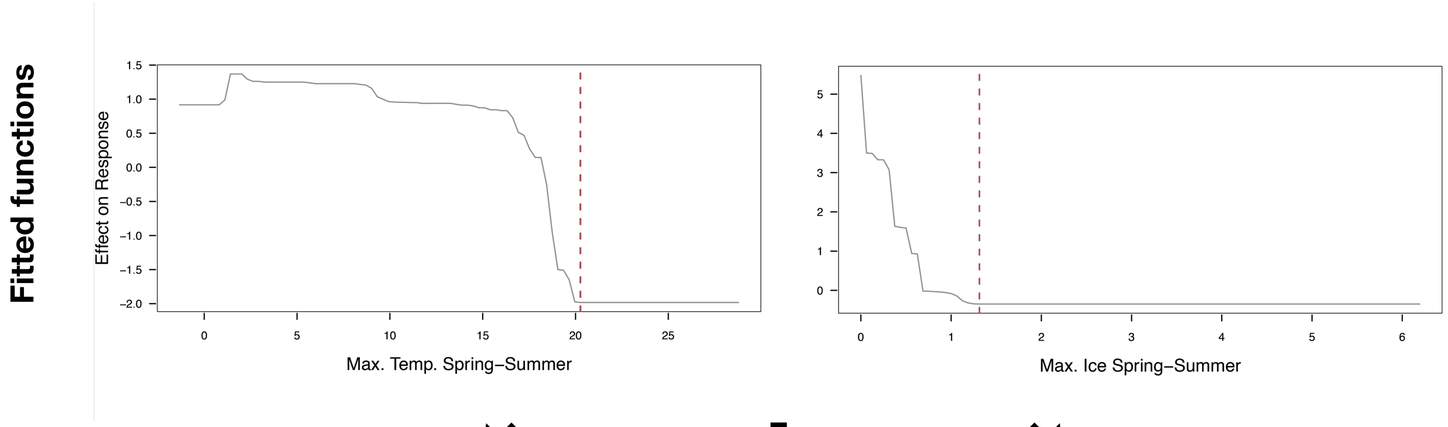
Predictive model-based transferability

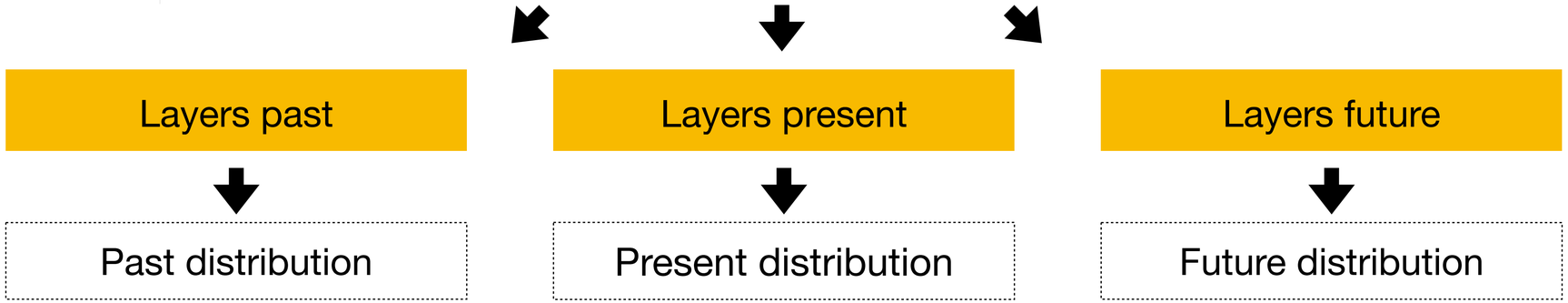
Ability of a model to predict beyond the data it was originally trained on.

Process: a model fits **observed species occurrence data against environmental variables for the corresponding period**. The **same model is then applied directly to new climate data** representing new environmental conditions.

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Predictive model-based transferability





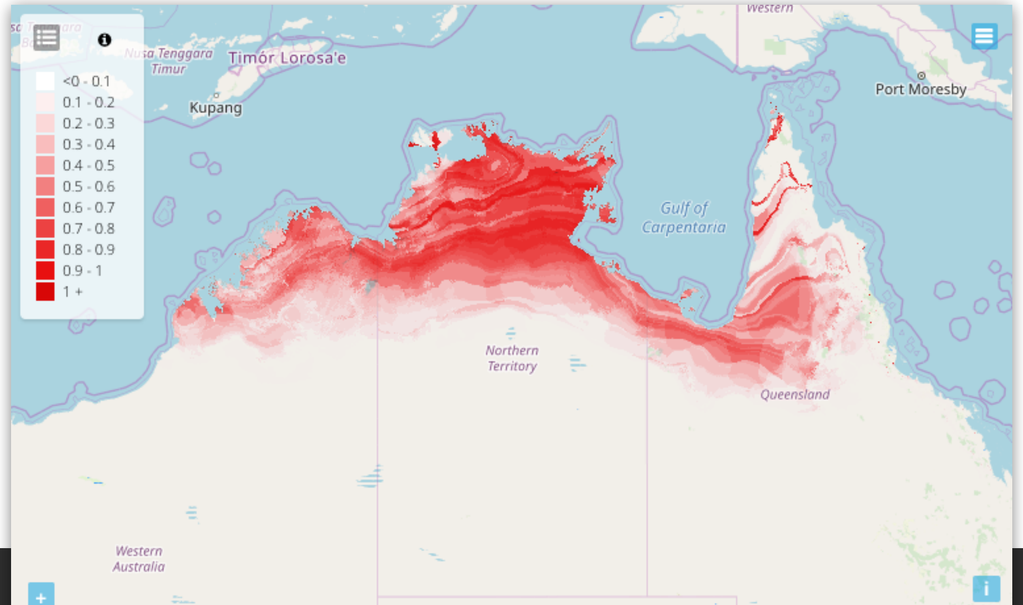
Regardless of the time period / region, **all environmental layers must be included in the transferability process**. So, the **availability of layers for climate scenarios also determines the choice of predictors for model fitting**.

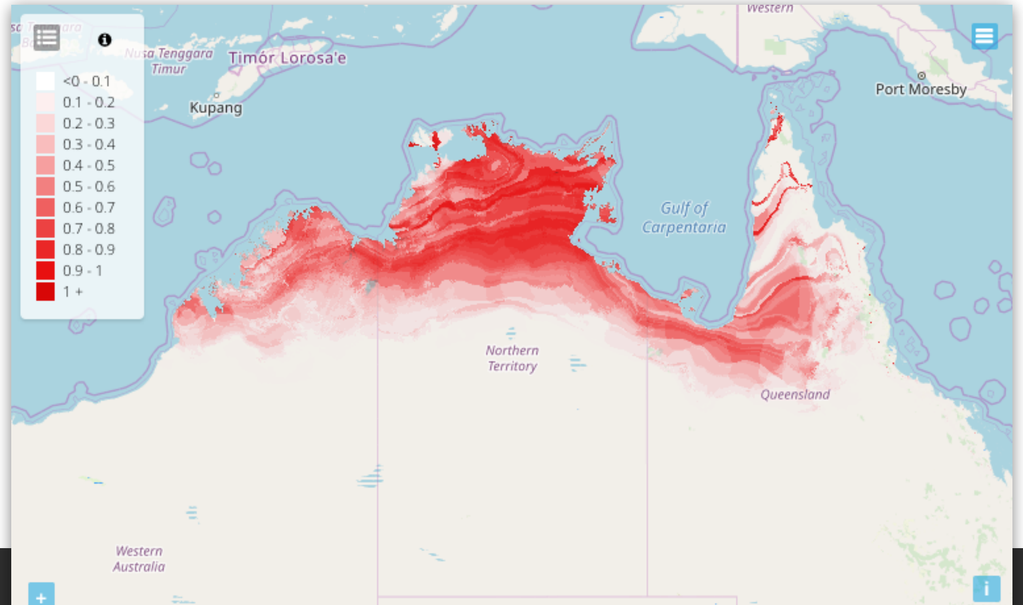
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Predictive model-based transferability

Model transferability can be analyzed (per cell) with different approaches:

(1) comparing **predictions** (e.g., probability) under different conditions.



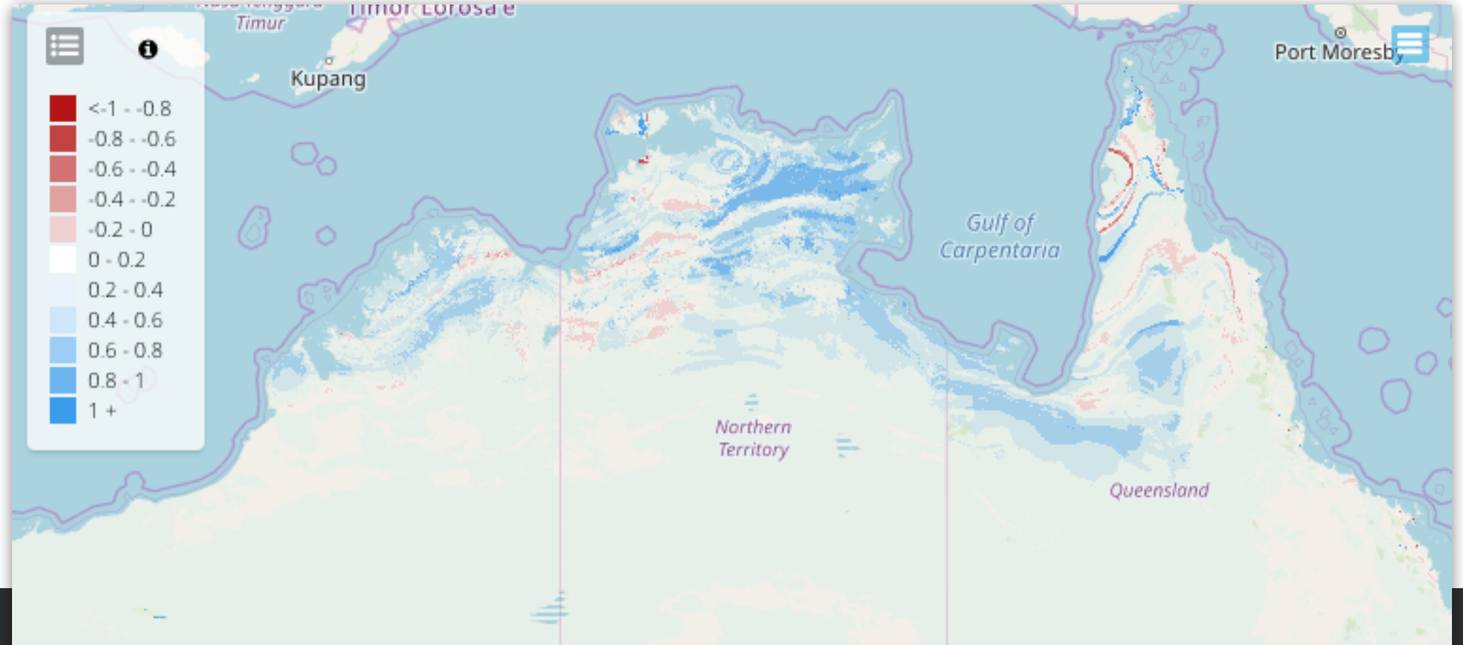


(2) **change in probability**, i.e., the difference in the predicted probabilities of different conditions;

e.g.,

[Layer 1 Future probability] **-** [Layer 2 present-day probability]

The map scales from -1 to 1, where negative values refer to lower suitable conditions and positive higher suitability.



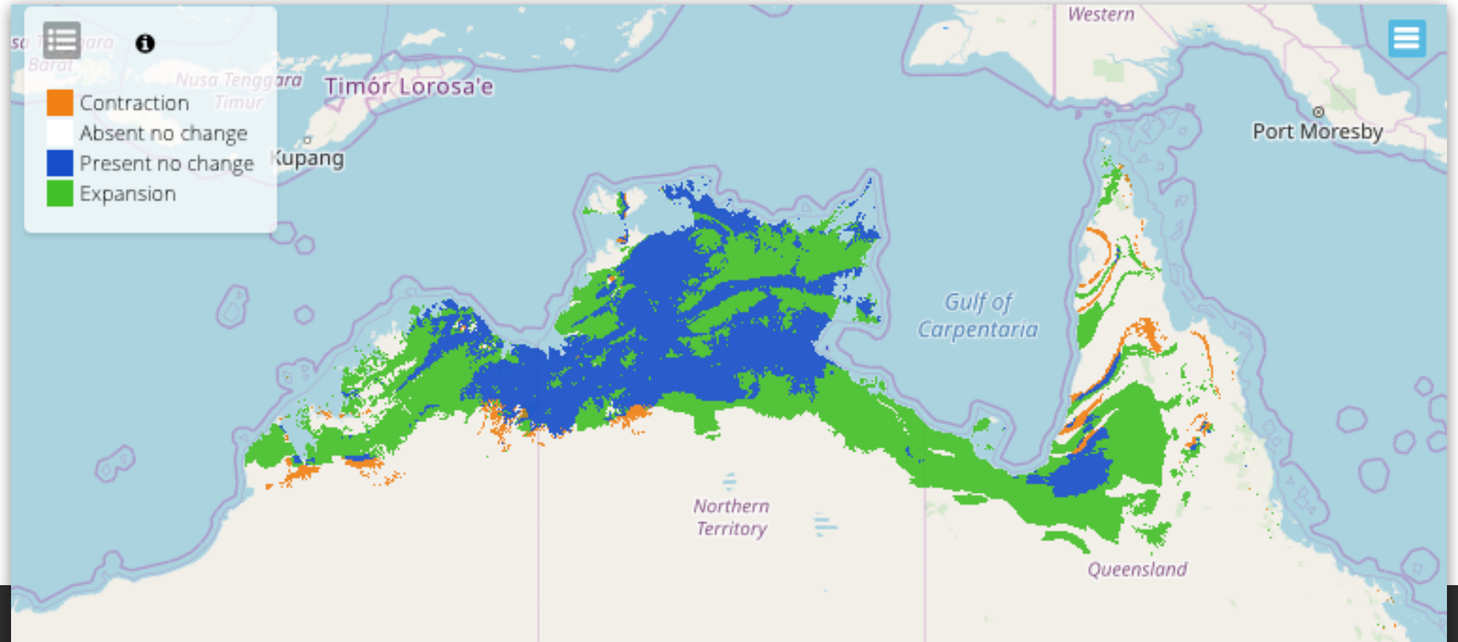
(3) **change in species distributional range**, generated with binary maps (i.e., 0 or 1).

Comparing maps can indicate:

No change: present distribution equals future distribution;

Range loss or contraction: presence in the present and absence in future;

Range expansion: absence in the present and presence in the future.



2 min. PITCH

Papers: P55

10 minutes

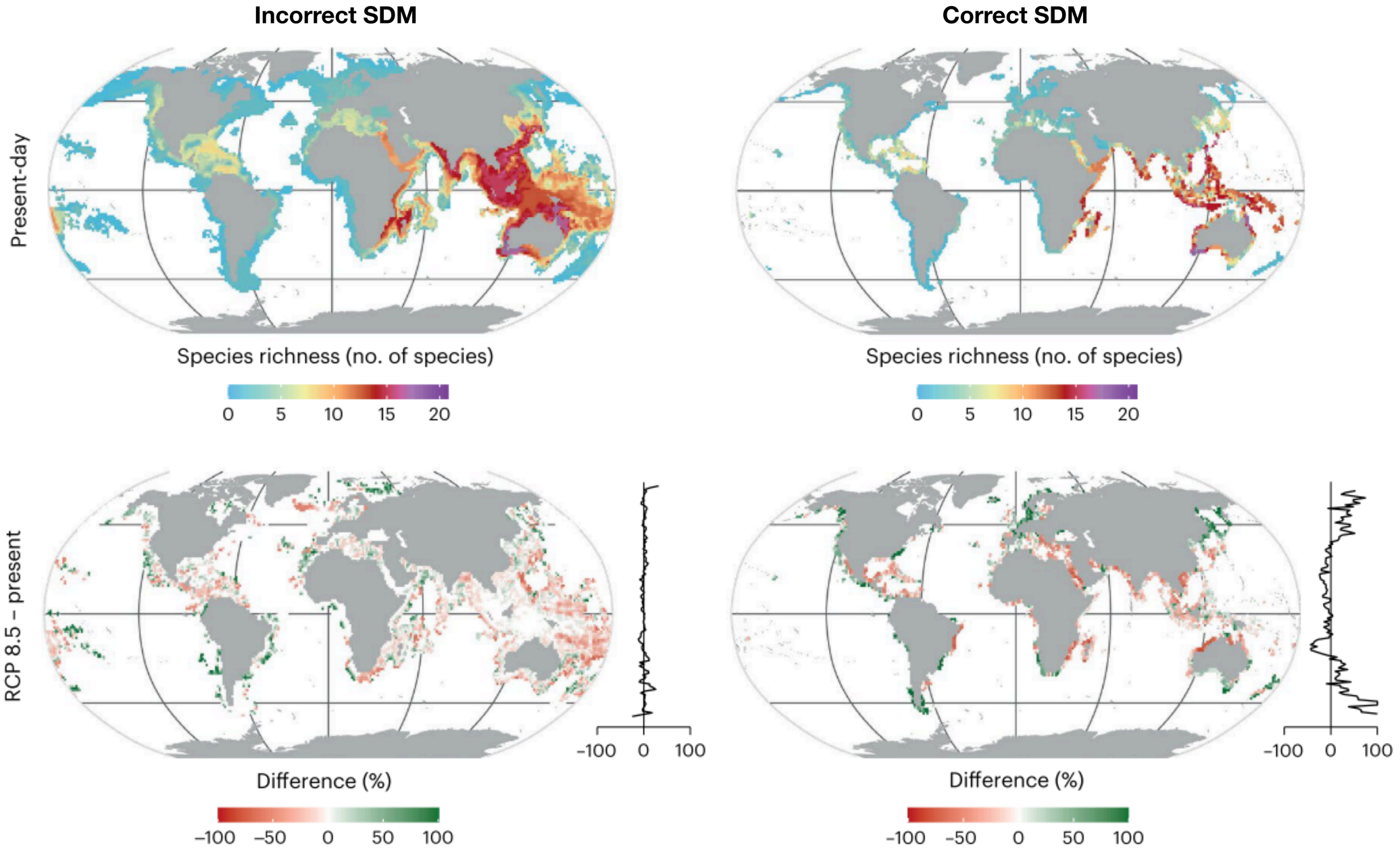
Read the abstract and the introduction and interpret the main figures.

Question. What problem did the authors tackle? (one sentence)

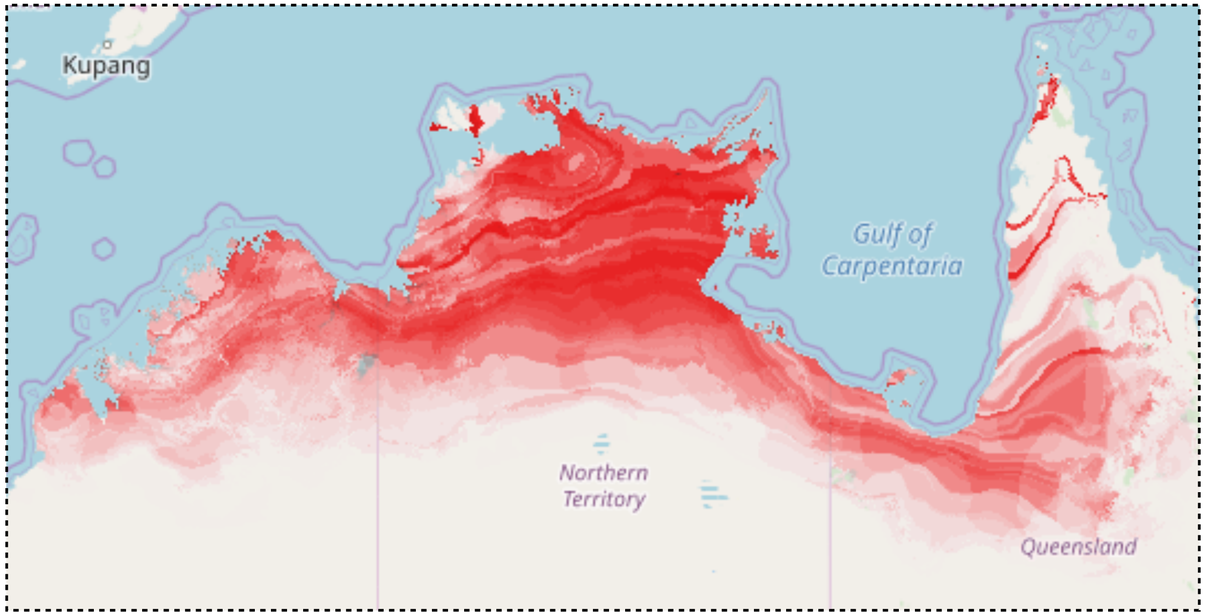
Approach. What did they do, and with which data and model? (one sentence)

Key finding. The one result to remember, ideally with the paper's main figure.

Why it matters today?



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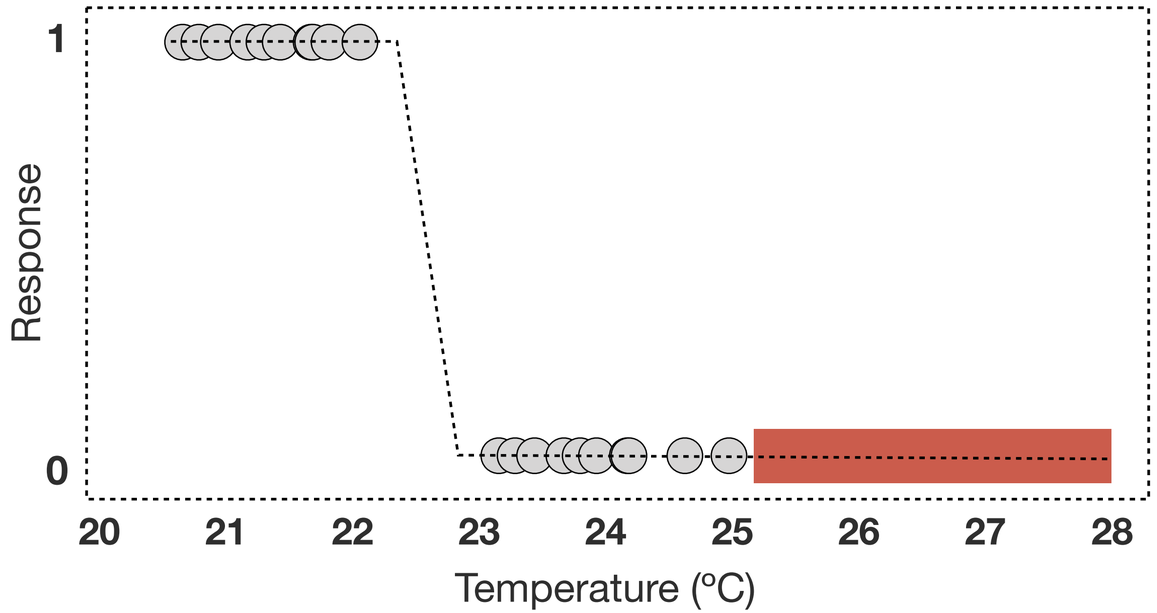


Predictive model-based interpolation

Predictions to **new sites within the range of values of environmental conditions sampled in the training data**.

From **scattered presence records to continuous distribution surfaces**.

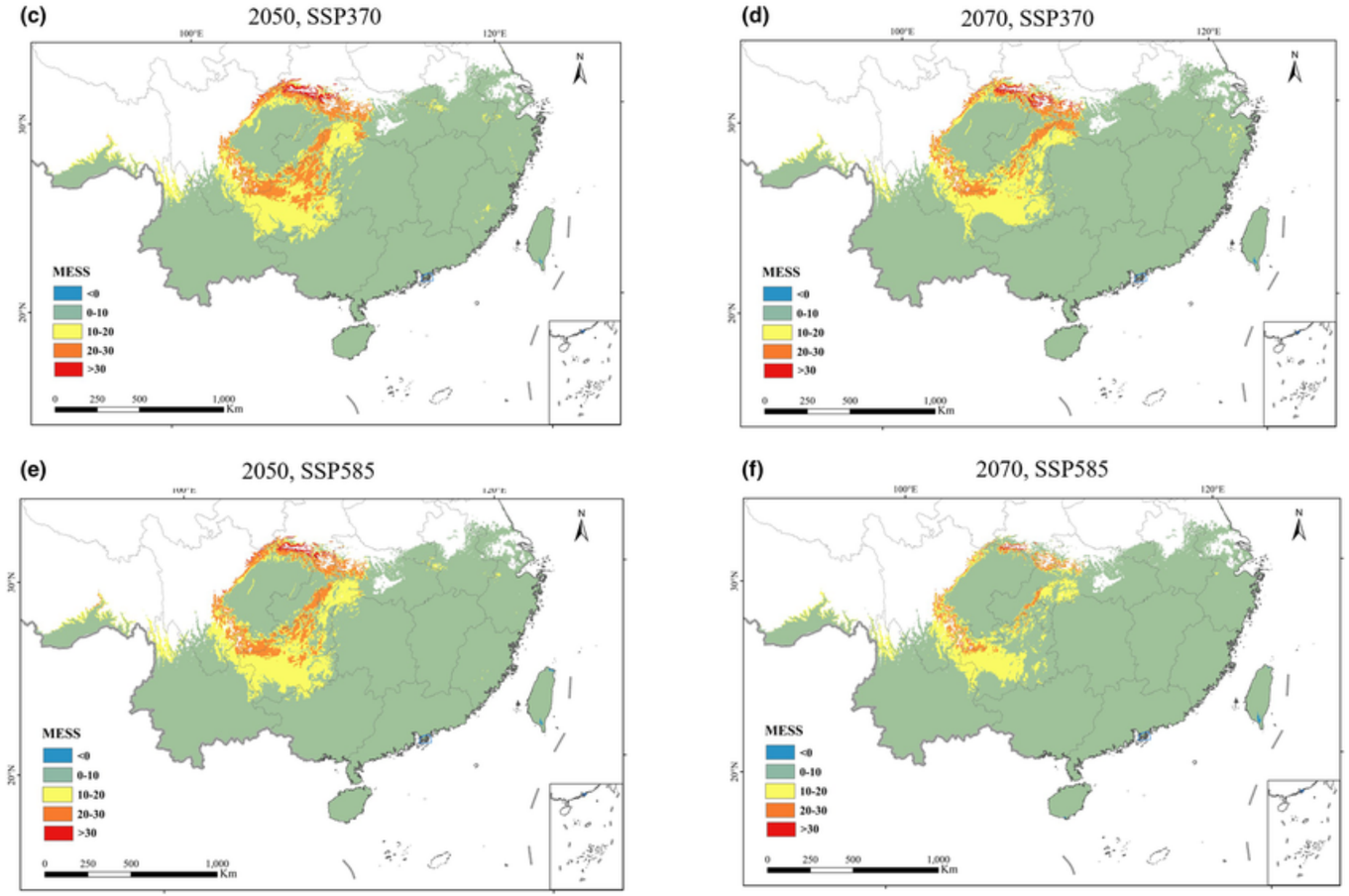
Maps do not show where species occur, but rather the distribution of regions with suitable environmental conditions.



Model transferability may lead to extrapolation, i.e., predictions to environmental values outside the range of environmental values used to fit the model.

e.g.,

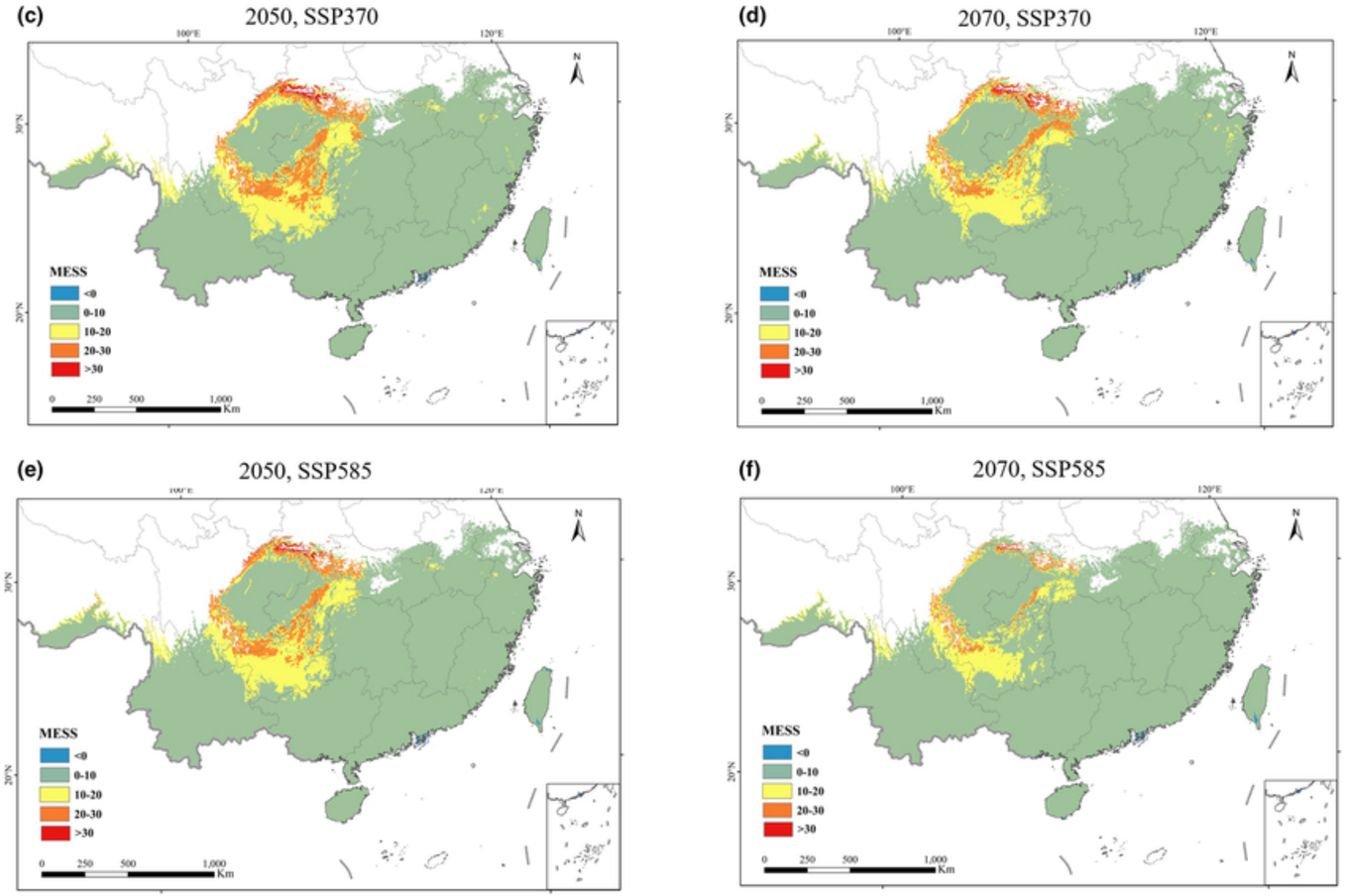
A **model fitted records with temperatures of 20-25ºC**. If **predictions** are **made for temperatures > 25ºC**, then the model will extrapolate. No information exists on the probability of occurrence at > 25ºC, so the **predictions will be uncertain**.



Quantifying Environmental Novelty

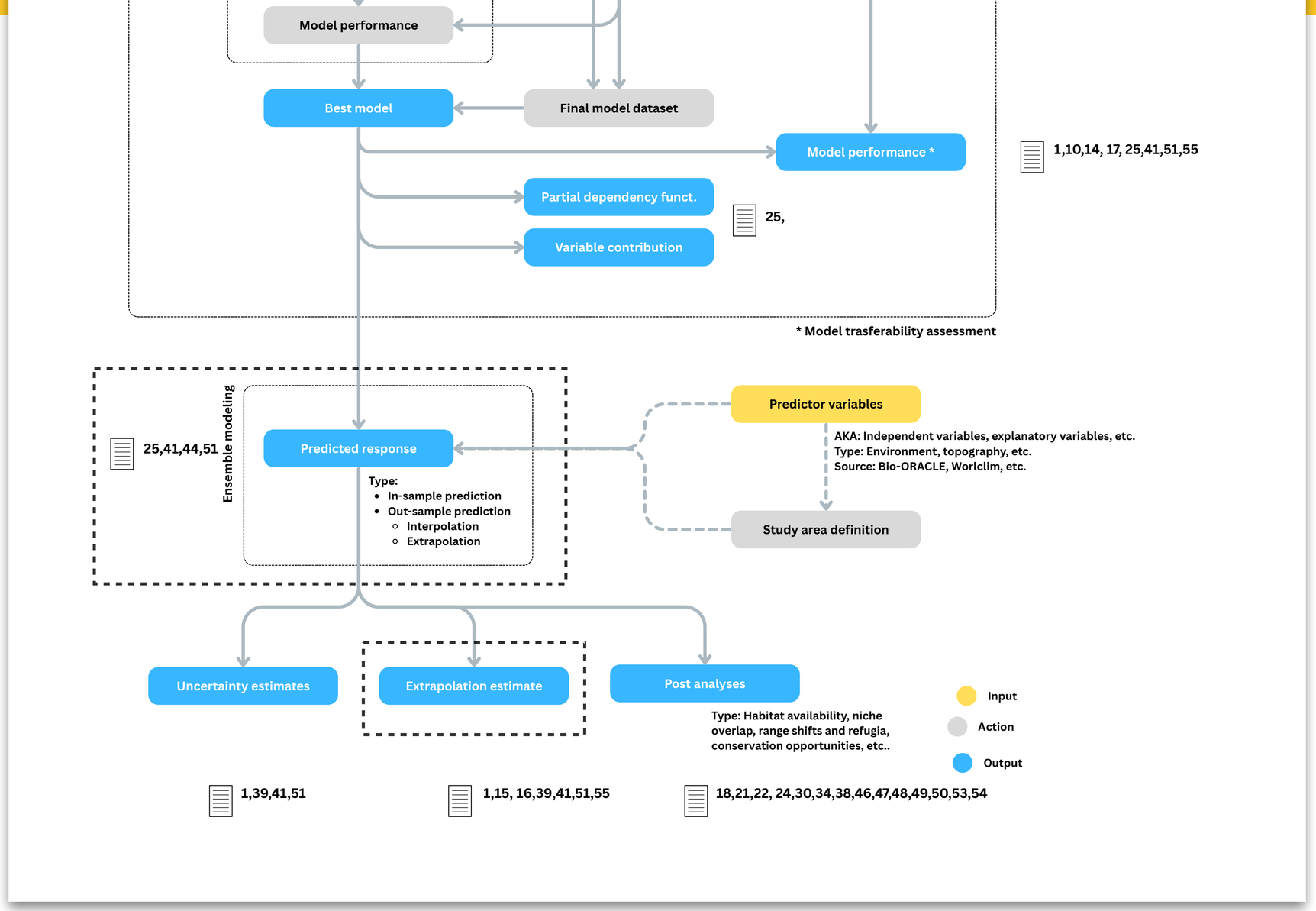
Future climates or new regions often contain conditions the model has never seen.

Multivariate Environmental Similarity Surfaces: A technique to map how environmentally similar (or dissimilar) the projection area is compared to the calibration area.



Positive Values (0 to 100): Similar Environments

Negative Values (< 0): Novel Environments (Extrapolation)


Model performance
Best model
Final model dataset
Model performance *
1,10,14, 17, 25,41,51,55
Partial dependency funct.
Variable contribution
25,
* Model trasferability assessment
Ensemble modeling
25,41,44,51
Predicted response
Type:
In-sample prediction
Out-sample prediction
Interpolation
Extrapolation
Predictor variables
AKA: Independent variables, explanatory variables, etc.
Type: Environment, topography, etc.
Source: Bio-ORACLE, Worlclim, etc.
Study area definition
Uncertainty estimates
Extrapolation estimate
Post analyses
Type: Habitat availability, niche overlap, range shifts and refugia, conservation opportunities, etc..
Input
Action
Output
1,39,41,51
1,15, 16,39,41,51,55
18,21,22, 24,30,34,38,46,47,48,49,50,53,54